

Peter Wang

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EDUCATION

University of California, Berkeley (GPA: 3.67)

May, 2026

B.S. in Computer Science & Minor in Electrical Engineering

- **Honors:** Jim & Donna Gray Endowment Award (\$10,000), Leadership Scholar (\$2,000), IEEE Member
- **Relevant Coursework:** Computer Architecture, Digital Design & Integrated Circuits, Microelectronic Devices and Circuits, Machine Learning, Probability & Random Processes, API Development, NLP.

SKILLS

- **Languages & Frameworks:** Verilog, Java, Python, C/C++, CUDA, Go, SQL, Matlab, JavaScript, Node.js, JUnit
- **Libraries:** PyTorch, Polars, NumPy, Pandas, SciPy, Matplotlib, Seaborn, Scikit-learn, Pydantic, SQLAlchemy
- **Tools:** Synopsys DC, Cadence Genus, VCS, Innovus, ModelSim, Vivado, OpenROAD, Yosys, GTKWave, SKY130 PDK

EXPERIENCE

Arm | SoC Architecture Intern

May 2025 — Aug. 2025

- Collaborated with physical design teams on the [Neoverse CSS](#) project to analyze pin counts and protocol mappings across IP instances for STA/timing closure.
- Built a Polars batch-inference pipeline to classify 20M+ pins and identify AMBA protocols (e.g. CHI, AXI, APB) in ~15 min; cut peak RAM by 85% for SoC-scale protocol analysis.
- Linked instance paths to die-level centroids coordinates that support across pre/post-restructuring formats.

Berkeley SLICE Lab | ML Accelerator Architecture Researcher

Jun. 2025 — May 2026

- Designing a flexible 2D systolic array for the [Zebra accelerator](#) to support operations beyond simple multiply-add, improving compute unit utilization with minimal control overhead.
- Implementing RTL and functional models to enable flexible compute units and efficient on-chip communication via a tightly coupled systolic mesh network.

UC Berkeley - College of Engineering | EECS Peer Advisor

Sep. 2024 — May 2025

- First peer advisor for EECS advising team, supporting 1,000+ students with academic planning ([eeecs/advising](#))

IEEE | Machine Learning Engineer

Aug. 2021 — Jun. 2024

- Published research paper "[Finger-Vein Recognition using a NASNet with a Cutout](#)" in 2021 ISPACS, IEEE.
- Designed Python ML models based on the NAS framework (Gradient-based Optimization).
- Utilized CUDA for parallelized training, implementing custom CUDA kernels to accelerate convolutional layers and matrix multiplications, reducing model training time by 36%.

Honda Research Institute | AI/ML Intern

Jan. 2024 — May 2024

- Tuned and applied L2 Ridge Regression to resolve multicollinearity in EV telematics data, enabling more accurate energy usage modeling.
- Clustered 100K+ driver sessions with K-means to uncover driving patterns and propose route adjustments, reducing urban congestion hotspots.
- Co-authored capstone research project: "[Potential of Telematics Data in Data Science](#)" (PDF)

Research & Hardware Projects

3-Stage RISC-V ASIC Design

- Designed and verified a 3-stage pipelined RISC-V CPU in Verilog, resolving pipeline hazards and implementing ALU/memory interfaces with custom testbenches.
- Synthesized Verilog RTL using SKY130 PDK, optimizing timing closure, die area, and power usage in a full RTL-to-GDS workflow.

AWS Trainium Accelerator | Kernel Optimization Project

- Optimized inference kernels using Neuron Kernel Interface (NKI) with engine-aware compute mapping and manual memory scheduling, achieving low-latency execution on AWS Trainium.

Branch Predictor Design

- Built a perceptron predictor using weighted history, reducing misprediction by ~13% (e.g., QSort: 0.176 to 0.153) in synthetic correlated-branch tests.